Ch 5.1.3-4: k-Fold Cross-Validation

Lecture 13 - CMSE 381

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Wed, Oct 4, 2023

Announcements

Last time:

- Validation Set
- LOOCV

Announcements:

- HW # 4 due Mon
-

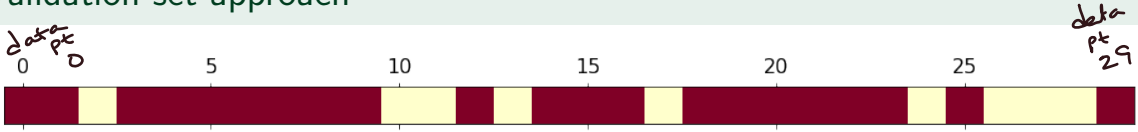
Covered in this lecture

- k-fold CV

Section 1

Last time

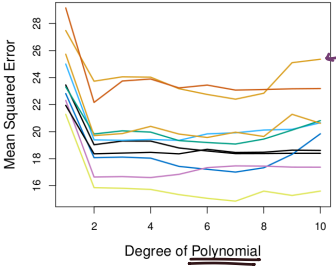
Validation set approach



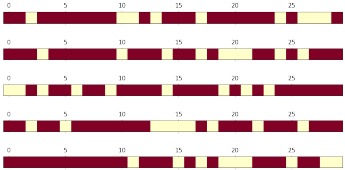
- Divide randomly into two parts:
  - Training set
  - Validation/Hold-out/Testing set
- Fit model on training set
- Use fitted model to predict response for observations in the test set
- Evaluate quality (e.g. MSE)

Yellow $\approx 30\%$
$\rightarrow$ Test set

Problems



one line = one chosen test set



Ex. Predict `mpg` using `horsepower`

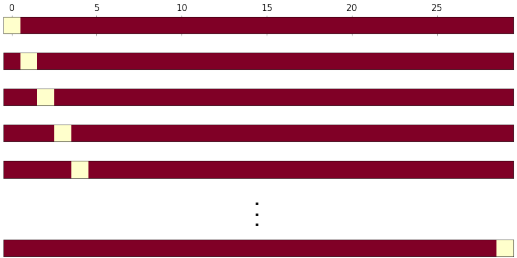
- Highly variable results, no consensus about the error
- Tends to overestimate test error rate

Leave One Out CV (LOOCV)

- Remove (x_1, y_1) for testing.
- Train the model on $n-1$ points: $\{(x_2, y_2), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_1 = (y_1 - \hat{y}_1)^2$

- Remove (x_2, y_2) for testing.
- Train the model on $n-1$ points: $\{(x_1, y_1), (x_3, y_3), \cdots, (x_n, y_n)\}$
- Calculate $\mathrm{MSE}_2 = (y_2 - \hat{y}_2)^2$

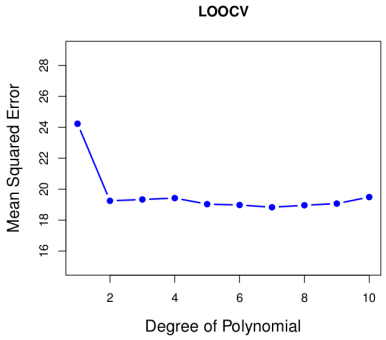
- Rinse and repeat



Return the score:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{MSE}_i$$

Pros and Cons



- No variance
- Higher computation cost

Speeding up LOOCV

Warning: This only works for least squares linear or polynomial regression.

$$h_i = \frac{1}{n} + \frac{(x_i - \overline{x})^2}{\sum_{j=1}^{n}(x_j - \overline{x})^2}$$

leverage statistic

$$\frac{1}{n}\sum_{i=1}^{n} \mathrm{MSE}_i = CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_i}\right)^2$$

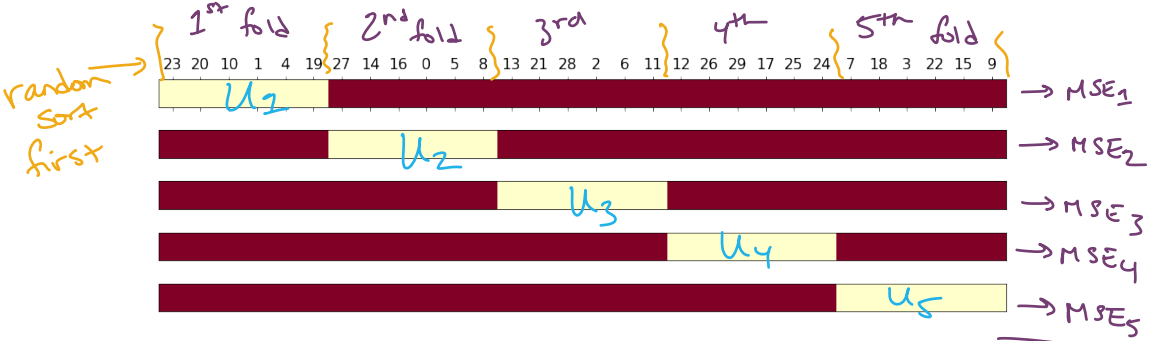
Learn model on all data

Section 2

k-Fold CV

The idea

Picture $k=5$ each collection is called a "fold"



Return the average

Note that if $K=n$ subsets

↳ This is LOOCV !!!

Mathy version

Random Sort

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} (y_j - \hat{y}_j)^2$

  # of test set data points
- Rinse and repeat

Return

$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{MSE}_i$$

averaging over all the folds

By hand first!

There are 10 students in the class, and we have data points for each. They have already been randomly permuted below. Write down the training/testing sets for a 3-fold CV

- Damien
- Alice
- Greta
- Jasmin
- Benji
- Inigo
- Firas
- Carina
- Enrique
- Hubert

U_1

U_2

U_3

Run ~~Fold~~ 1	Run ~~Fold~~ 2	Run ~~Fold~~ 3
Training		
BIFCEH (everything else)	DAGJCEH	DAGJBIF
Test		
U_1	U_2	U_3
DAGJ	BIF	CEH

Coding - Building k-fold CV

$j=0$

$0 \quad 1\cdot l \quad l+1 \quad 2l \quad 2l+1 \quad 3l \quad 3l+1 \quad \ldots \quad kl$ (k)

$n // k \quad l$

[0 1 2 3 4 5 6 7 8 9 10 11]

$j\cdot l$

$0\cdot l \quad 1\cdot l \quad 2\cdot l \quad 3\cdot l \quad 4\cdot l$

list$[jl : (j+1)\,l]$

$j=0 \quad 0 : 1\cdot l$

$j=1 \quad 1\cdot l : 2\cdot l$

$j=2 \quad 2\cdot l : 3\cdot l$

Pros and Cons $\boxed{\text{k-fold}}$

Pros:

Cheaper computation than LOOCV

train model k times

↳ often k = 5, 10

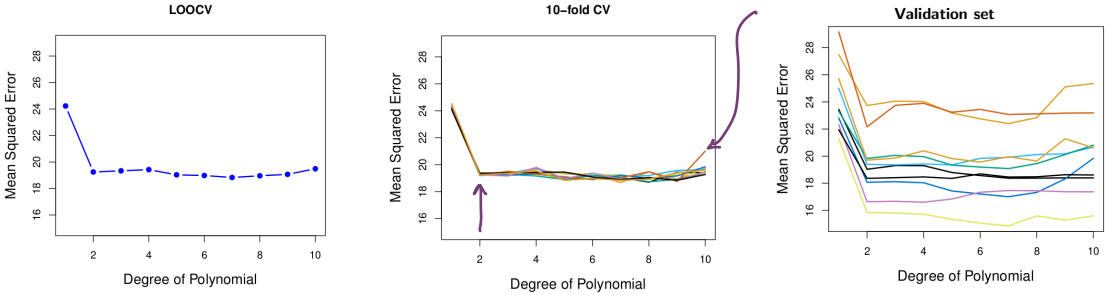
Cons:

There is randomness (unlike LOOCV)

but variance is lower than validation set version

Comparison

each line = shuffle data



What polynomial degree should I use?

Next time

Lec #	Date			Reading	Homeworks
12	Mon	Oct 2	Leave one out CV	5.1.1, 5.1.2	
13	Wed	Oct 4	k-fold CV	5.1.3	
14	Fri	Oct 6	More k-fold CV,	5.1.4-5	Quiz
15	Mon	Oct 9	k-fold CV for classification	5.1.5	HW #4
16	Wed	Oct 11	Resampling methods: Bootstrap	5.2	
17	Fri	Oct 13	Subset selection	6.1	
18	Mon	Oct 16	Shrinkage: Ridge	6.2.1	
19	Wed	Oct 18	Shrinkage: Lasso	6.2.2	
	Fri	Oct 20	**Review**		
	Mon	Oct 23	No class - Fall break		
	Wed	Oct 25	**Midterm #2**		
20	Fri	Oct 27	Dimension Reduction	6.3	